# The Knop et al. (2009) Non-Monotonic Formal Solver in CMFGEN

An undergraduate-level explanation of why the original solver broke, and how the new one works

SN 2024pba modelling notes

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**Abstract**

CMFGEN computes supernova spectra by solving radiative transfer in the *comoving frame*, the frame of reference that rides along with the expanding gas. Its original formal solver assumed that the gas velocity always increases outward ($dv/dr > 0$). The STELLA hydrodynamic snapshots used for SN 2024pba violate this: the SN_B shell ploughing into the wind creates compression zones where the velocity momentarily *decreases* with radius ($dv/dr < 0$). In those zones the old solver produced negative optical depths, asked the wrong neighbouring frequency for photons, and crashed or silently returned wrong answers. This document explains, at an undergraduate level, how the original solver worked, why it failed, and how the new solver, based on the formal solution of Knop et al. (2009), fixes it by separating the *strength* of the velocity coupling (kept in the opacity, always positive) from its *direction* (kept in the source function, where the sign decides which neighbouring frequency a photon came from).

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# 1 The two problems CMFGEN has to solve

A supernova spectrum is set by a tug-of-war between two sets of equations, and CMFGEN (Hillier & Miller, 1998) solves them together:

1. **Radiative transfer.** How much light of each frequency $\nu$ travels in each direction through the gas? The answer is the *specific intensity* $I_\nu$, which obeys

$$\frac{dI_\nu}{ds} = \eta_\nu - \chi_\nu I_\nu, \tag{1}$$

   where $s$ is distance along a light ray, $\eta_\nu$ is the *emissivity* (new light created per unit length) and $\chi_\nu$ is the *opacity* (fraction of light removed per unit length). The ratio $S_\nu = \eta_\nu/\chi_\nu$ is the *source function*: the intensity the gas "tries" to reach.
2. **Statistical equilibrium.** How many atoms sit in each energy level of each ion? Those *populations* $n_i$ decide $\eta_\nu$ and $\chi_\nu$, but they themselves depend on the radiation field through radiative excitation and ionisation rates proportional to the mean intensity $J_\nu$ (the direction-averaged $I_\nu$).

Neither can be solved first, so CMFGEN iterates: guess populations → solve transfer to get $J_\nu$ → solve the rate equations to get better populations → repeat until nothing changes ($\Lambda$-iteration, accelerated by Ng extrapolation (Ng, 1974) and approximate-Newton "BA" corrections). The step that computes $I_\nu$ and $J_\nu$ for fixed populations is called the *formal solution*, and it is exactly that step that the non-monotonic solver replaces. Everything else in the iteration cycle is untouched.

# 2 Why expansion complicates things: the comoving frame

Equation (1) looks simple, but the ejecta is expanding at up to $10^4\,\mathrm{km\,s^{-1}}$. A photon travelling outward through differentially moving gas is constantly Doppler-shifted: in the frame of the gas it started next to, its frequency steadily drops. CMFGEN therefore works in the *comoving frame* (CMF), the frame moving with the local gas, where absorption lines sit at fixed rest frequencies $\nu_0$, but a photon's CMF frequency *marches* along the ray.

## 2.1 One number summarises the velocity gradient: $\sigma$

CMFGEN packages the local velocity gradient into

$$\sigma \;=\; \frac{d\ln v}{d\ln r} - 1 \;=\; \frac{r}{v}\frac{dv}{dr} - 1. \tag{2}$$

Homologous expansion ($v \propto r$) gives $\sigma = 0$. The frequency drift rate along a ray is then

$$\Gamma \;=\; \frac{v}{r}\left(1 + \sigma\mu^2\right), \tag{3}$$

with $\mu$ the cosine of the ray angle. In ordinary expansion $\Gamma > 0$: photons redshift toward lower CMF frequencies as they travel outward. Where the gas *decelerates* with radius (a compression zone, e.g. behind a shock), $\sigma < -1$ and $\Gamma$ flips sign: photons *blueshift* instead. That sign flip is the entire source of trouble, as the next two sections show.

# 3 How the original solver worked

## 3.1 Folding the frequency drift into a fake opacity

The full CMF transfer equation contains an extra term describing the frequency drift of $I_\nu$. The original CMFGEN solver absorbed that term into a *generalised opacity*,

$$\hat{\chi} \;=\; \chi \;+\; \frac{\Gamma}{\Delta\ln\nu}, \tag{4}$$

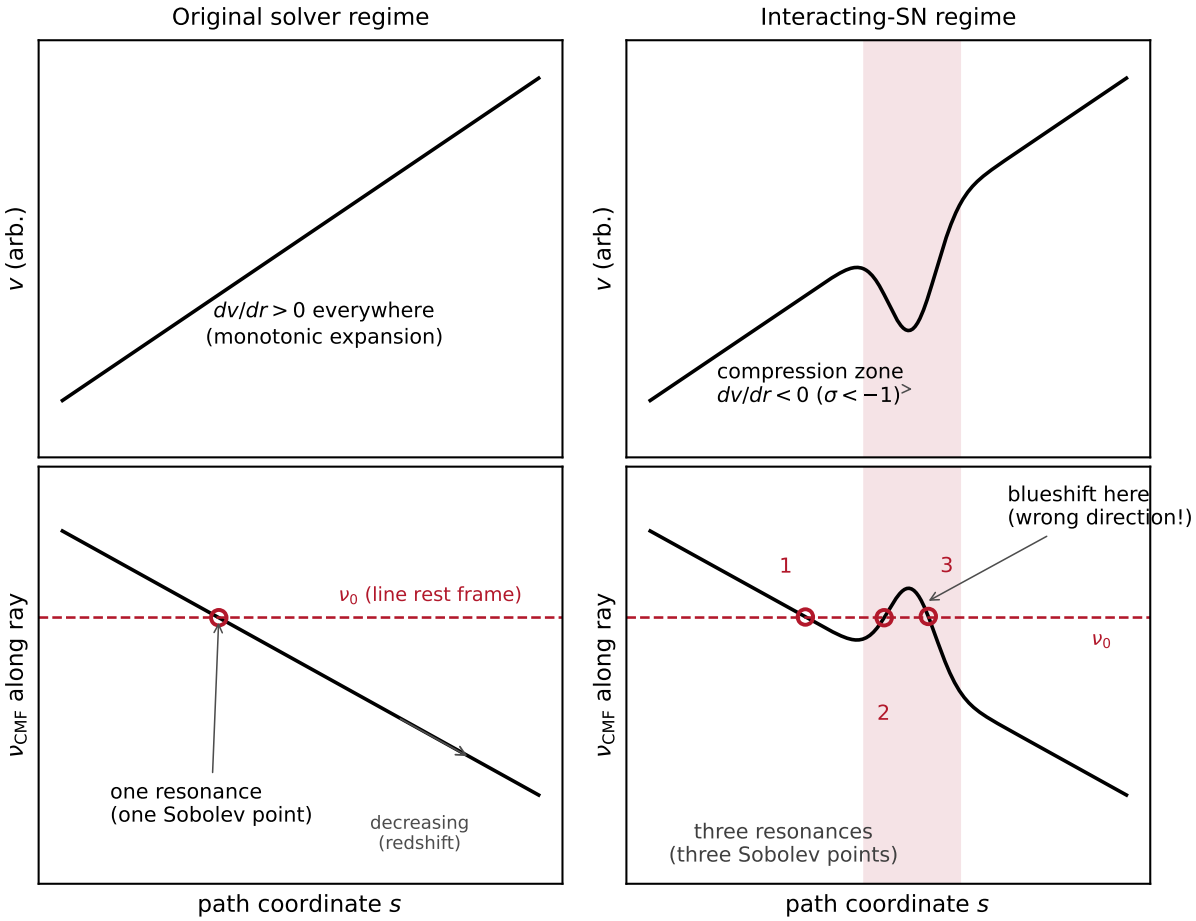



Figure 1: Why monotonic velocity matters. *Top*: velocity profiles. *Bottom*: the comoving-frame frequency a fixed observer-frequency photon experiences as it travels along a ray. In a monotonic outflow (left) the CMF frequency decreases smoothly and crosses each line rest frequency $\nu_0$ exactly once. In a compression zone (right, shaded) the photon is *blueshifted*: $\nu_{\rm CMF}$ turns around and crosses $\nu_0$ three times, creating three resonance points the original solver did not know how to handle.

where $\Delta \ln \nu$ is the spacing of the frequency grid. The transfer equation then looks like the static one,

$$\frac{dI}{d\hat{\tau}} = I - S', \qquad d\hat{\tau} = \hat{\chi}\, dz, \qquad S' = S + Q\, I_{\rm upwind}, \qquad Q = \frac{\Gamma/\Delta \ln \nu}{\hat{\chi}}, \tag{5}$$

which is solved along each ray by *short characteristics*: start at a boundary where $I$ is known, and step along the ray using the neighbouring grid points already computed.

### 3.2 The sweep order and the upwind direction

CMFGEN walks the frequency grid from high $\nu$ to low $\nu$. Because expanding gas only ever redshifts photons, a photon at frequency $\nu$ can *only* have arrived from a *higher* frequency, the point already computed in the sweep. That previously computed intensity is $I_{\rm upwind}$ in Eq. (5): it is always the *previous* frequency point, and it is always available. Figure 1 (left column) shows why this works: with $dv/dr > 0$ everywhere, the CMF frequency along any ray decreases monotonically, so each spectral line at rest frequency $\nu_0$ is crossed *exactly once*: one Sobolev (resonance) point per ray.

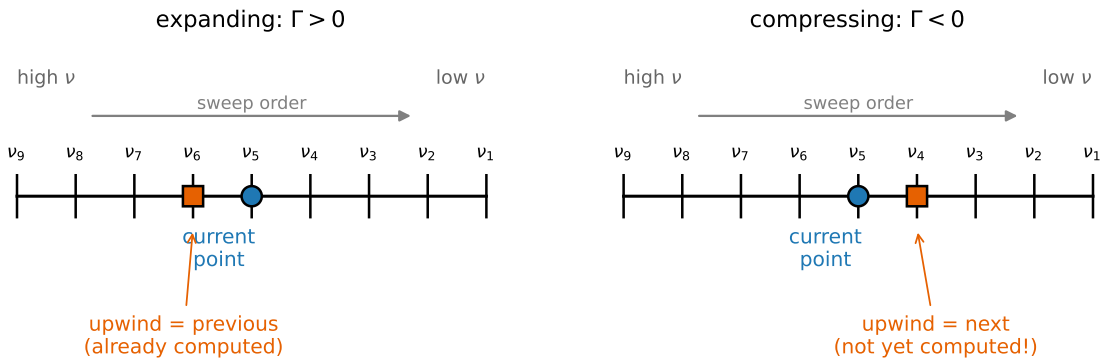



Figure 2: The upwind-frequency choice on CMFGEN's frequency grid (sweeping high $\rightarrow$ low $\nu$). In expanding gas ($\Gamma > 0$) photons redshift, so the photons at the current point came from the *previous*, higher frequency, already computed. In a compression zone ($\Gamma < 0$) photons blueshift: they came from the *next*, lower frequency, not yet computed in this sweep. The Knop solver uses it anyway, from the previous *iteration*'s stored values.

### 3.3 Why monotonicity was safe

Two assumptions are baked into Eqs. (4) and (5):

1. $\Gamma > 0$ everywhere, so $\hat{\chi} > \chi > 0$: the generalised opacity is positive, $\hat{\tau}$ is positive, and the solution factor $e^{-\hat{\tau}}$ attenuates light as physics demands.
2. Photons only ever come from the previous (higher) frequency point, which the sweep has already computed.

For a smooth, monotonically expanding atmosphere both hold automatically, which is why the solver worked for decades.

## 4 What a compression zone does to the original solver

A STELLA snapshot of an interacting supernova contains regions where the fast shell material decelerates into the slower medium ahead, a compression zone with $dv/dr < 0$, hence $\sigma < -1$ and $\Gamma < 0$. Three things then go wrong at once:

1. **Negative optical depth.** In Eq. (4) the term $\Gamma/\Delta \ln \nu$ goes negative and can make $\hat{\chi} < 0$. A negative opacity means $e^{-\hat{\tau}}$ *amplifies* light instead of attenuating it. Intensities grow exponentially, go negative, and NaNs spread through the arrays.
2. **Photons arrive from the wrong direction.** Where the gas compresses, photons are blueshifted as they travel, so the photons arriving at frequency $\nu$ came from a *lower* frequency: the *next* point in the sweep, which has not been computed yet. The solver nevertheless keeps using the previous point and gets the wrong answer even where it does not crash.
3. **Multiple resonances.** As Fig. 1 (right) shows, a single observer-frame frequency can resonate with the same line at *three* different locations along a ray. A solver built for one Sobolev point per line per ray cannot represent this; and in the observer-frame transfer the bookkeeping index that walks the frequency grid could be sent in the wrong direction entirely (in the SN 2024pba model the old one-directional index search was wrong at $\sim$99.5% of ray points).

The old code already "knew" this could happen and protected itself by *clamping* $\sigma \geq -0.999$ before the formal solver saw it. That prevented crashes, but it erased the compression zone from the physics, which is precisely the feature we are trying to model. It also produced subtle wrongness rather than honest failure.

# 5 The Knop et al. (2009) idea: separate strength from direction

Knop et al. (2009) developed a formal solution for radiative transfer in non-monotonic velocity fields (in the context of PHOENIX/3-D supernova transfer). The version implemented here follows the same logic. The central trick is to stop bundling two different things into one signed number:

- the *strength* of the velocity coupling (how rapidly photons drift in frequency) belongs in the opacity, and opacity must be positive;
- the *direction* of the coupling (which neighbouring frequency photons arrive from) belongs in the source term, where a sign is perfectly legal.

## 5.1 The new equations

Equation (4) is replaced by

$$\hat{\chi} = \chi + \frac{|\Gamma|}{\Delta \ln \nu} \qquad \text{(magnitude only: always } > 0\text{)} \tag{6}$$

$$Q = \frac{\Gamma/\Delta \ln \nu}{\hat{\chi}} \qquad \text{(signed: keeps the direction)} \tag{7}$$

$$S' = S + Q\, I_{\text{upwind}}, \tag{8}$$

with the upwind neighbour chosen by the *sign* of $\Gamma$:

$$I_{\text{upwind}} = \begin{cases} I_{\text{PREV}} & \Gamma \geq 0 \quad \text{(expanding, as before)} \\ I_{\text{NEXT}} & \Gamma < 0 \quad \text{(compressing: Knop fix).} \end{cases} \tag{9}$$

The differential equation actually integrated, $dI/d\hat{\tau} = I - S'$, is unchanged in form, so all the careful short-characteristics machinery still applies. What changed is that $\hat{\tau}$ can no longer be negative, and the source function knows which way the "river" of frequency coupling flows.

## 5.2 The catch: the next frequency hasn't been computed yet

CMFGEN sweeps from high to low frequency, so at the current point $I_{\text{NEXT}}$ (the lower-frequency neighbour) does not yet exist in this sweep (Fig. 2). The Knop scheme handles this with a *Gauss–Seidel iteration across "great iterations"*, the same philosophy as the $\Lambda$-iteration CMFGEN already uses:

1. On the first great iteration, set $I_{\text{NEXT}} = 0$ at compressing points. Crude, but stable: those points just get $S' = S$ (no frequency coupling) for one round.
2. After each great iteration, store the computed intensities.
3. On the next great iteration, use the stored values as $I_{\text{NEXT}}$.
4. Repeat until the stored and computed intensities agree.

## 5.3 Multiple resonance crossings

Because $\nu_{\text{CMF}}$ can turn around along a ray (Fig. 1), one observer-frame frequency can hit the same line at several depths. The new solver integrates the ray *piecewise*: it marches along the ray, detects every sign change of the frequency drift, and applies the short-characteristics step segment by segment, so each resonance crossing is handled with the correct local direction. In the observer-frame module the bracketing index $K$ is now allowed to move *either way* along the frequency grid (decrement first, then increment), rather than only ever increasing. This small change fixed the $\sim$99.5% index failure rate on the SN 2024pba model.

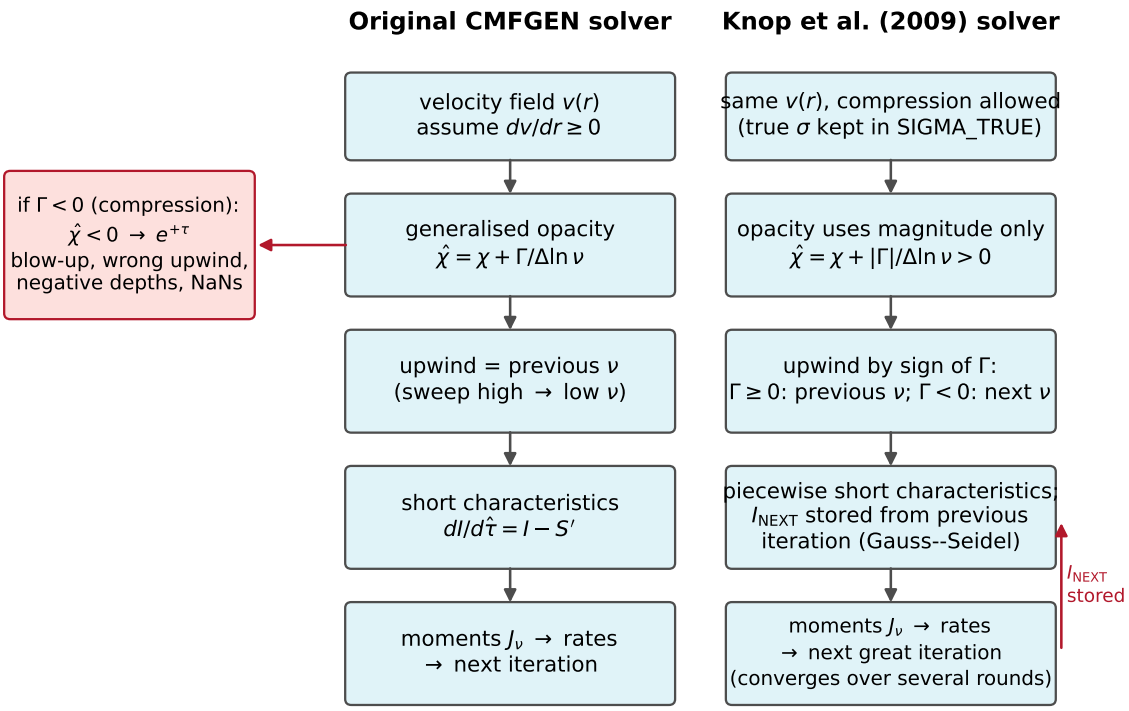



Figure 3: The two formal solvers side by side. Left: the original path, which implicitly requires $\Gamma > 0$ (monotonic expansion) and fails in compression zones. Right: the Knop et al. (2009) path: the opacity carries only $|\Gamma|$, the signed coupling $Q$ lives in the source function, the upwind neighbour is chosen by the sign of $\Gamma$, and the not-yet-computed $I_{\rm NEXT}$ is supplied from the previous great iteration.

### 5.4 Two $\sigma$ arrays: telling the truth selectively

Much of CMFGEN's surrounding machinery (the moment equations, the BA acceleration matrix, Ng extrapolation, Sobolev escape probabilities, regridding) was written assuming $\sigma > -1$ and is unstable or meaningless otherwise. Rather than rewrite all of it, the solver keeps two arrays:

| Array | Contents | Who sees it |
|---|---|---|
| `SIGMA_TRUE` | true, unclamped $d\ln v/d\ln r - 1$ | the Knop formal solver only |
| `SIGMA` | clamped, $\geq -0.999$ | every other routine (stability) |

The formal solution, the place where the physics of the compression actually enters the spectrum, sees the real velocity field. The legacy plumbing sees a safe, fake-monotonic version and keeps working. A useful analogy is a city map: the delivery driver (the formal solver) gets the true map with all the one-way reversals; the traffic-planning spreadsheet (the moment equations) gets a simplified map that never reverses, so its assumptions still hold.

Because the moment equations cannot be trusted at $\sigma < -1$, the non-monotonic path bypasses them: the Eddington factors that close the system are computed directly from the ray moments (`GET_MOMS_NON_REL`, selected by `USE_LAM_ES=T`) instead of from a tridiagonal moment solve.

## 6 A river analogy

For intuition, think of the frequency grid as a river flowing through a row of towns (grid points), with photons as leaves on the water:

- **Normal flow** (expansion): the current carries leaves downstream, from high frequency to low. Each town only needs to ask its upstream neighbour (already visited) how many leaves to expect.
- **Reversed stretch** (compression zone): the current runs *backwards*. Two bugs appear: the old formula counted the current's strength *with its sign*, so a reversed current made the "muddiness" (opacity) negative, so leaves would multiply as they drift. And the town now needs leaf counts from *downstream*, a town not yet visited this round.
- **The Knop fix**: make the muddiness depend on the current's speed $|\Gamma|$, never its sign. The river is always muddy. Keep the sign separately to know which neighbour to ask. For the not-yet-visited downstream town, remember what it said *last* round; over several rounds the remembered answers converge to the truth.

## 7 What actually changed in the code

The solver is opt-in (`NON_MONO_SOLVER=T` in `VADAT`); a monotonic model is completely unaffected. The main pieces are:

| Piece | Role |
|---|---|
| `FG_J_CMF_NONMONO_V1` | The new formal solver: Eqs. (6)–(9), piecewise integration across multiple crossings, Gauss–Seidel storage of $I_{\rm NEXT}$. Replaces `FG_J_CMF_V13` when enabled. |
| `SIGMA_TRUE` | Parallel array carrying the unclamped velocity gradient; filled wherever the velocity law is set (hydro input, restart, observer module) and linearly interpolated to the ray grid. |
| Bidirectional $K$-index (`OBS_FRAME_SUB_V9`) | Observer-frame bracketing index may now decrement *or* increment, because in a compression zone the transformed frequency can move either way along the grid. |
| `USE_LAM_ES=T`, `GET_MOMS_NON_REL` | Bypass the (unstable for $\sigma < -1$) moment-equation solve; take Eddington factors straight from the ray moments. |
| Forced `VADAT` settings | `GLOBAL_LINE=BLANK`, `DO_SOB_LINES=F`, `VEL_LAW=11`, `PURE_HUB=F` etc. are enforced so the non-monotonic path is self-consistent. |

A full routine-by-routine description lives in `docs/cmfgen_and_nonmono_solver.tex` and `docs/CMFGEN_NONMONO_CHANGES.md`; this document is the pedagogical version.

## 8 Why SN 2024pba needs this

In our scenario the STELLA snapshot handed to CMFGEN contains a dense SN_B shell driving a forward shock at the shell/wind interface, with radiatively heated (but unshocked) SN_A ejecta beyond it. The velocity field across that interface is non-monotonic: the shocked gas decelerates into the slower wind, producing exactly the compression zones ($\sigma < -1$) that break the original solver. Without the Knop formal solution the options were both bad: crash, or clamp $\sigma$ and silently erase the shock structure from the radiative transfer. With it, CMFGEN computes the spectrum on the *actual* STELLA velocity field: the shocks and compression zones survive into the formal solution.

Two honest caveats remain. First, stability is not accuracy: the BA Jacobian still uses the clamped $\sigma$, so convergence in strongly non-monotonic regions is approximate and slower. Second, the Gauss–Seidel coupling of $I_{\rm NEXT}$ needs several great iterations to converge, so non-monotonic runs need more iterations before their spectra are trustworthy.

## References

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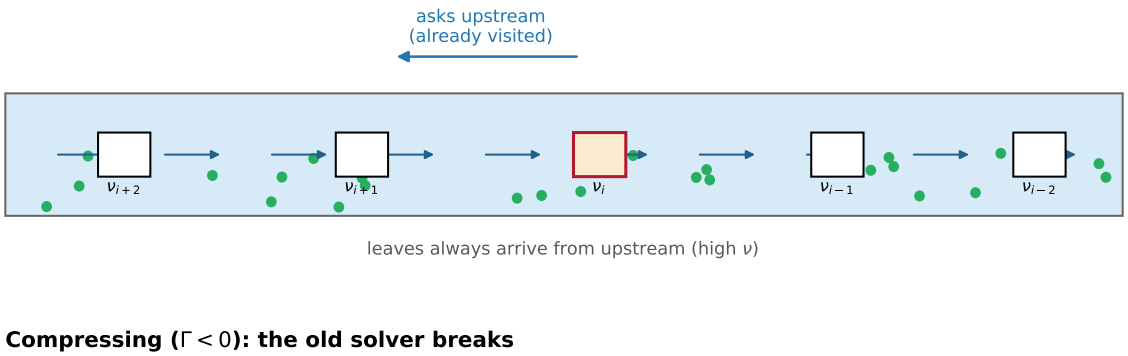



Figure 4: The river analogy of §6, drawn. Each panel is a stretch of the frequency grid: towns are grid points, leaves are photons, the current is the velocity coupling Γ, and the muddiness of the water is the opacity $\hat{\chi}$. *Top:* in an expanding flow the current runs one way (high $\nu$ to low $\nu$, left to right), so each town asks the upstream neighbour it has already visited. *Middle:* in a compression zone the current reverses; the old solver still asks upstream and gets the wrong leaf count, while the signed coupling also makes the muddiness negative. *Bottom:* the Knop solver asks downstream, using the previous iteration's stored intensities, and the muddiness depends only on $|\Gamma|$.